

Teaching A Reinforcement Learning Agent to Dance

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Abstract

Dance learning requires the integration of rhythm, movement, memory, and feedback. In this project, we used reinforcement learning to model how an agent can learn a fixed dance-like sequence through repeated trial and error. We represented choreography as a 32-move target sequence composed of eight possible movements and organized around an eight-count rhythmic structure. Using Q-learning, we compared different reward structures, including a random baseline, sparse reward, and chunked/curriculum learning, as well as different state representations, including memorization, beat-only, and beat + previous action models. Performance was measured using the longest correct prefix of the target choreography, with full mastery defined as correctly reproducing all 32 movements. The chunked/curriculum reward structure supported learning more effectively than sparse reward, suggesting that intermediate feedback was important for mastering long sequences. In the state representation comparison, the memorization model performed best because it encoded exact sequence position, while beat-only and beat + previous action models lacked enough information to uniquely identify each target movement. Overall, these results show that reward structure and state representation strongly influence an agent's ability to learn choreography. However, the model simplifies dance into a fixed target-matching task and does not fully capture the creativity, imitation, embodiment, or aesthetic judgment involved in real human dance learning.

Introduction

Dance is an intrinsic part of being human. The synchronization between movement and sound in dance has been linked in the past to pleasure (Zentner and Eerola, 2010). Dance also operates at the intersection of motion and rhythm. Although dance can exist without music, it is often organized through rhythm, timing, and patterned movement. Prior work has shown that dance engages many brain regions, such as in Basso et al, 2021. In particular, dance affects numerous areas (Figure 1), including the

prefrontal cortex, motor areas, somatosensory cortex, temporal areas, parietal areas, cingulate, caudate nucleus, etc. (Basso et al, 2021).

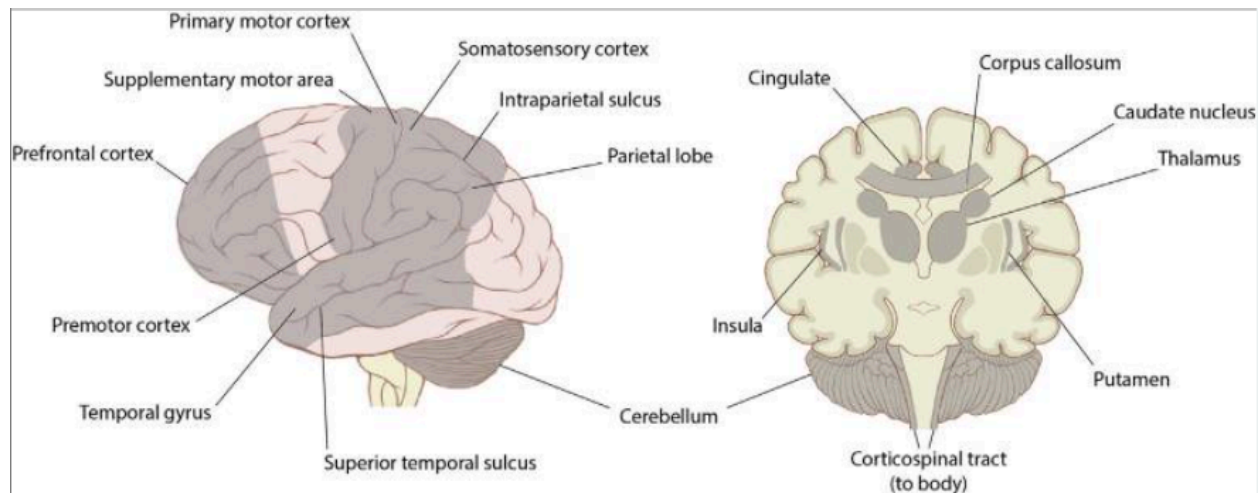


Figure 1: Image from Basso et al, 2021 & Patrick J. Lynch *via* Creative Commons. **Left: Lateral view of the brain with regions shown to be affected by dance highlighted in gray. Right: Coronal section of the brain with regions shown to be affected by dance highlighted in gray.**

Dance also appears to have deep developmental ties to rhythm, as studies show that infants can move in synchrony with musical rhythms (Phillips-Silver et al., 2010). This synchronization may be mediated by corticostriatal loops, neural feedback circuits connecting cortical motor regions with the basal ganglia that support sequential and motor learning (Carol A. Seger & Earl K. Miller, 2010). According to Seger and Miller, these loops enable the brain to gradually transform repeated sensory and motor experiences into generalized movement patterns through reinforcement and feedback. In the context of dance, repeated exposure to rhythm and choreography may allow dancers to develop automatic motor “chunks,” in which complex movement sequences become integrated into fluid and coordinated actions. Jordi Fonollosa et al. (2015) suggest that chunking emerges through hierarchical neural dynamics that organize sequential actions into stable patterns. In dance, repeated rehearsal may allow individual movements to become grouped into coordinated motor chunks that can later be executed automatically and fluidly. This reduces cognitive load and improves timing, precision, and synchronization with music. The pauses that often occur between choreographic phrases may reflect transitions between these motor

chunks, similar to the delays observed at chunk boundaries in computational and behavioral studies of sequence learning.

Dance is also related to reinforcement learning because the latter is not only a computational framework for sequential decision-making, but also has strong parallels in biological systems underlying sequence learning. In studies of habit formation in macaques, striatal neurons in the caudate nucleus have been shown to encode integrated cost-benefit signals during the learning of action sequences (Desrochers 2015). This suggests that sequential behaviors can be shaped by reinforcement-driven optimization over extended action chains, rather than isolated decisions. More broadly, RL frameworks model behavior through an internal “state” representation, which serves as a compressed summary of relevant environmental and task information that guides future actions (Langdon 2020). This notion of state is particularly relevant for structured sequence generation tasks such as choreography, where latent structures including rhythm, motifs, and transitions must be maintained over time rather than inferred from single-step decisions.

The goal of this project was to test whether reinforcement learning could learn a short dance-like movement sequence using a simplified eight-count structure. Specifically, we trained agents to reproduce a predefined target choreography, where each beat corresponded to an objectively correct movement. We compared how different state representations affected learning, including models that used only the beat position and models that also incorporated the previous movement as contextual information. We evaluated performance based on how accurately the learned sequence matched the target choreography and how reward accumulated over training.

Methods

To model choreography computationally, we represented a short dance phrase as a fixed 32-move target sequence built from movements labeled A–H. The task was not to generate an original choreography from aesthetic principles, but to learn a predefined sequence of movements through reinforcement. Each episode required the agent to attempt the sequence by selecting one movement at a

time. After each action, the agent received feedback according to the reward structure being tested, allowing us to compare how different forms of feedback shaped learning.

Beat	1	2	3	4	5	6	7	8
Move	A	C	D	F	B	G	H	A

Figure 2: Example Structure of an eight-count. This eight-count structure was used as a simplified representation of rhythmic position in choreography. In this setup, the agent selected a movement at each beat, generating a sequence of steps that together formed a short choreography. Depending on the model, the state encoded the current beat, the sequence position, and/or the previous movement. The current movement was selected by the agent as its action, rather than being given as part of the state.

We first compared three reward structures: a random baseline, a sparse reward model, and a chunked/curriculum model. The random baseline selected movements without learning and was used to estimate performance by chance. In the sparse reward condition, the agent only received meaningful reward if it completed the full 32-move target sequence correctly, making the task difficult because partial progress did not provide feedback. In the chunked/curriculum condition, the agent learned the choreography in smaller four-move chunks and advanced only after reaching a mastery threshold. This reward structure was intended to model the way dancers often learn choreography gradually, receiving feedback on smaller sections before integrating them into a full routine.

After identifying chunked/curriculum learning as the most reliable reward structure, we compared three state representations: memorization, beat-only, and beat + previous action. The memorization model encoded the agent’s position in the full sequence, allowing it to directly associate each sequence index with the correct movement. The beat-only model encoded only the current beat within an eight-count cycle, testing whether rhythmic position alone was sufficient for learning the choreography. The beat + previous action model encoded both the current beat and the previous movement, allowing the same beat to be treated differently depending on the immediately preceding action.

All learning models were implemented using Q-learning, a model-free reinforcement learning algorithm that learns the expected value of taking a given action in a given state. At each step of an episode, the agent observed its current state, selected one of the eight possible movements using an

epsilon-greedy policy, received reward feedback, and updated its Q-table. Early in training, the agent explored by selecting actions randomly with high probability. As training progressed, epsilon decayed, causing the agent to increasingly exploit the action with the highest learned Q-value for each state. This allowed us to compare how quickly each model learned the target choreography under different exploration schedules.

Each model was trained for 5,000 episodes. The initial exploration rate was set to $\epsilon = 1.0$, meaning the agent began by choosing actions randomly. The learning rate was set to $\alpha = 0.1$, and the discount factor was set to $\gamma = 0.9$. We tested epsilon decay values of 50, 100, 250, 500, and 1000 to examine how the rate of transition from exploration to exploitation affected learning.

Importantly, the movement selected on the current step was not included in the state before action selection. Instead, the state representation provided contextual information, such as the agent's position in the sequence, the beat within the eight-count, or the previous movement depending on the model. The agent then selected the current movement as its action. Thus, in the beat + previous action model, the same beat paired with different previous movements was treated as a distinct state, but the correct current movement was not given to the model in advance.

Performance was measured using the longest correct prefix of the target choreography. This metric counted the number of consecutive movements selected correctly from the beginning of the sequence before the first error. A score of 32 indicated full mastery of the target choreography. We tracked this value across training episodes to compare learning speed, final accuracy, and stability across reward structures, state representations, and epsilon decay rates.

Results

We first compared three reward structures to determine which type of feedback best supported learning of the target choreography: a random baseline, a sparse reward model, and a chunked/curriculum model. The random baseline selected movements without updating a Q-table, so its performance represented chance-level sequence matching. The sparse reward model used Q-learning, but only received meaningful reward when the entire 32-move sequence was completed correctly. In contrast, the

chunked/curriculum model received feedback while learning smaller four-move sections of the choreography before progressing to later chunks.

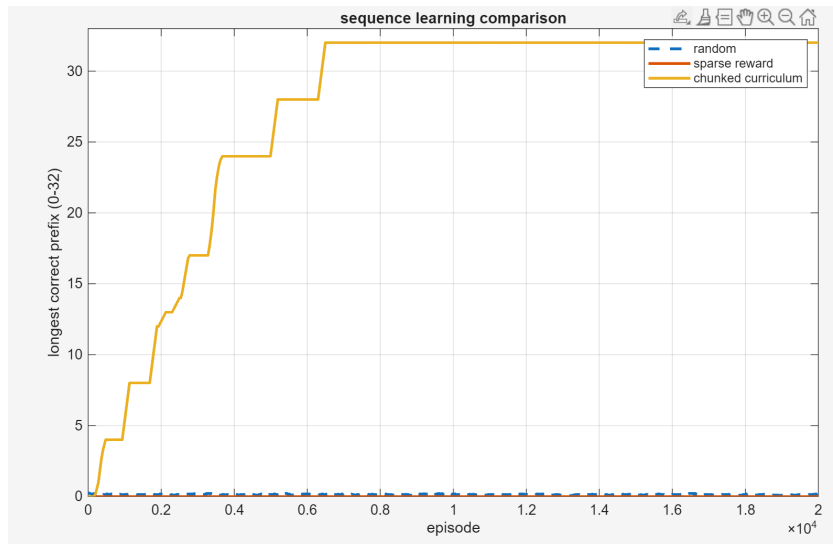


Figure 3: Comparison of reward structures across training. Longest correct prefix across episodes for the random baseline, sparse reward model, and chunked/curriculum model.

The reward comparison showed that the chunked/curriculum model learned the target choreography much more effectively than the other reward structures. Both the random baseline and sparse reward model remained near zero across training, indicating that they rarely produced a long correct prefix. By contrast, the chunked/curriculum model showed steady improvement and eventually reached the full 32-move sequence. This suggests that intermediate feedback was important for learning the choreography, because requiring the full sequence to be correct before providing reward made the task too difficult for the agent to learn efficiently.

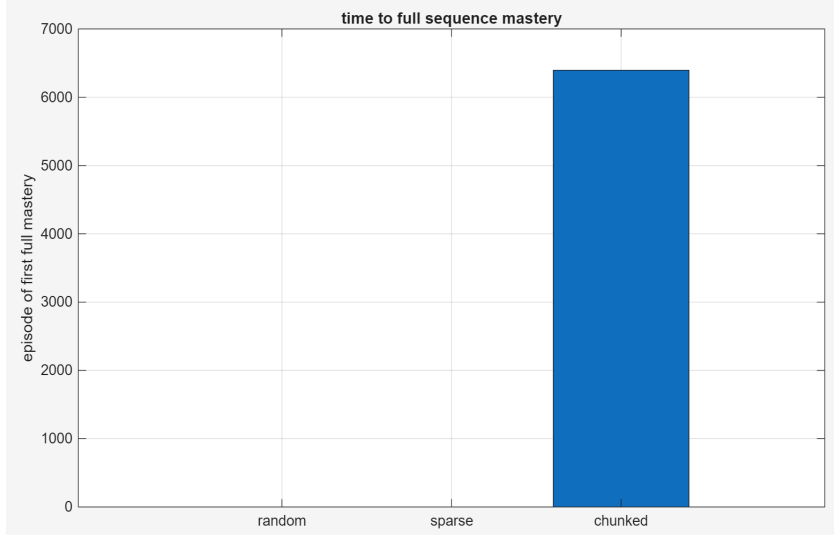


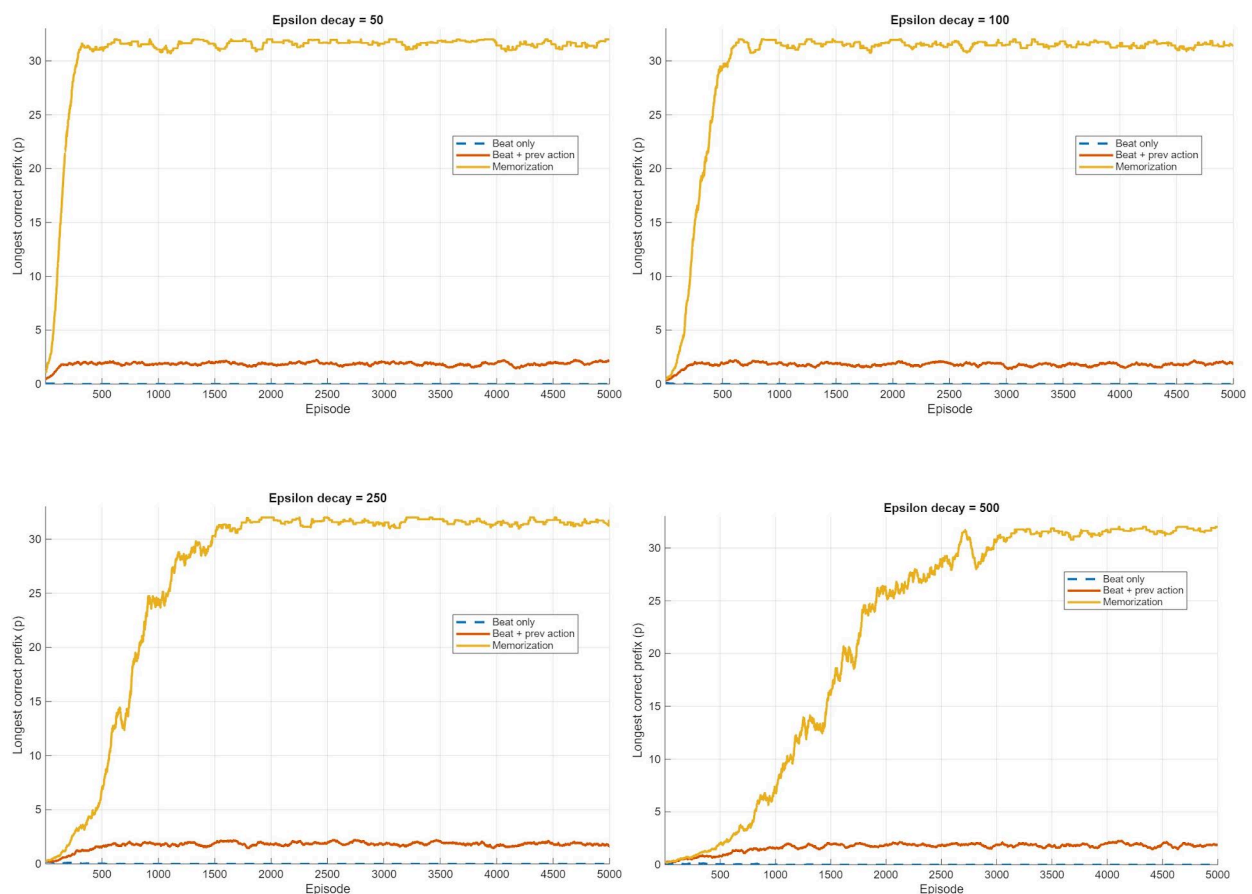
Figure 4: Time to full sequence mastery across reward structures. Number of episodes required for each reward structure to reach the full 32-move target sequence.

The time-to-mastery comparison further supported this result. The chunked/curriculum model was the only reward structure that reliably reached full sequence mastery. This indicates that dividing the choreography into smaller sections reduced the difficulty of the learning problem and allowed the agent to build up the sequence gradually. In this way, the chunked/curriculum approach more closely resembled how human dancers often learn choreography by mastering shorter phrases before combining them into a complete routine.

After identifying chunked/curriculum learning as the most reliable reward structure, we compared three state representations: memorization, beat-only, and beat + previous action. In all models, the agent’s action was the movement selected at the current step from the eight possible movements A–H. The action was therefore not included in the state before the agent made its choice. Instead, the state provided contextual information that the agent could use to choose an action. In the memorization model, the state encoded the agent’s exact position in the 32-move sequence. In the beat-only model, the state encoded only the current beat within the eight-count cycle. In the beat + previous action model, the state encoded both the current beat and the movement selected on the previous step. Therefore, the same beat paired

with different previous actions was treated as a different state. This allowed the model to use local movement history as context while still requiring the agent to choose the current movement through Q-learning.

We next compared how these state representations affected learning. The memorization model performed best because it encoded the agent's exact position in the sequence, allowing each position to be directly associated with the correct target movement. The beat-only model performed poorly because the same beat appeared multiple times across the 32-move choreography, so beat position alone was not enough to identify the correct movement. The beat + previous action model added local context, but still did not match memorization, suggesting that previous-movement information alone was insufficient to recover the full target sequence.



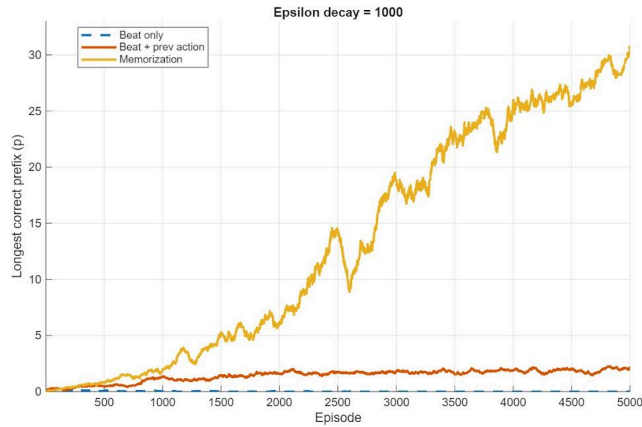


Figure 6: Model performance with epsilon decay = 50, 100, 250, 500, and 1000. Longest correct prefix across training episodes for the memorization, beat-only, and beat + previous action models.

Across epsilon decay values, the memorization model consistently outperformed the beat-only and beat + previous action models. Faster decay values allowed the memorization model to exploit learned state-action associations earlier, leading to faster convergence. Slower decay values delayed mastery because the agent continued exploring for more episodes. However, extended exploration did not substantially improve the beat-only or beat + previous action models, likely because their state representations did not contain enough information to uniquely identify every target movement.

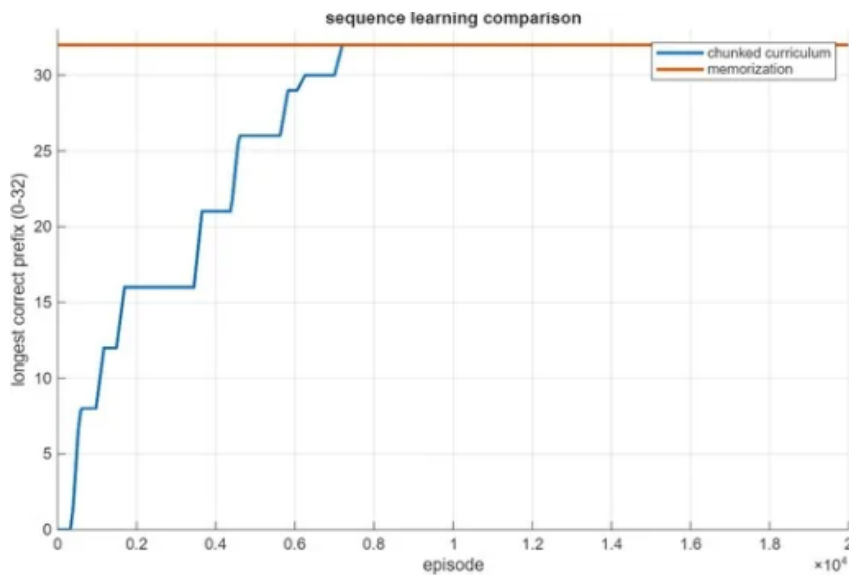


Figure 7: Sequence learning comparison between the chunked/curriculum and memorization models. Comparison of the two strongest approaches for learning the 32-move target choreography.

Finally, we compared the two strongest approaches: chunked/curriculum learning and memorization. Both methods were able to achieve strong performance, but they did so in different ways. The memorization model learned quickly because it encoded exact sequence position, making the correct movement easier to associate with each state. The chunked/curriculum model learned more gradually by breaking the choreography into smaller sections, but this approach more closely resembled how dancers often learn routines in practice. Overall, these results show that both reward structure and state representation strongly shaped the agent's ability to learn choreography. However, because the strongest models depended on either explicit sequence position or a fixed curriculum, they were optimized for reproducing a specific target sequence rather than generating flexible or novel choreography.

Conclusion

Overall, this project showed that a Q-learning agent could learn a fixed dance-like movement sequence when given an appropriate reward structure and state representation. Sparse reward alone was not sufficient for reliable learning, because the agent received feedback only after completing the entire 32-move sequence correctly. In contrast, chunked/curriculum learning allowed the agent to receive feedback on smaller sections of the choreography, making it easier to build toward full-sequence mastery. This suggests that intermediate feedback is important for learning long movement sequences, which parallels the way human dancers often learn choreography by mastering shorter phrases before combining them into a full routine.

The model comparison also showed that state representation strongly shaped learning. The memorization model performed best because it encoded exact sequence position, allowing the agent to directly associate each point in the choreography with the correct movement. The beat-only and beat + previous action models performed worse because they did not contain enough information to uniquely identify every movement in the fixed target sequence. This result suggests that, in our simplified model, knowing where the agent was in the sequence was more useful than knowing only the rhythmic beat or the immediately preceding movement.

However, this model does not fully reflect the reality of human dance learning or choreography. In many real dance classes, students learn by observing and imitating an instructor, which may be closer to supervised learning than reinforcement learning. Human dancers also receive rich feedback from many sources, including visual demonstration, verbal correction, proprioception, social cues, musical interpretation, and aesthetic judgment. In addition, dance is not only a memorized motor sequence, but also a creative practice. Wiggins et al. (2015) describe creativity as a process involving the generation and evaluation of novel possibilities within a structured domain. Our model reduced choreography learning to a target-matching task in which there was one objectively correct movement at each step. While this made the problem computationally tractable, it does not capture the full creative and embodied nature of dance.

This distinction is important because our project was not modeling how dancers create original choreography. Instead, it modeled a narrower question: how an agent can learn to reproduce a fixed movement sequence through repeated feedback. In this context, reinforcement learning was useful because the agent had to make a sequence of actions, receive reward feedback, and update future choices based on the consequences of previous attempts. However, the model's success depended heavily on simplifying dance into discrete actions, fixed beats, and a single correct target sequence. As a result, the strongest models were also the least flexible: they learned the target choreography effectively, but they were not capable of generating novel or creative dances.

Future work could address these limitations by expanding the model beyond target-sequence memorization. A more realistic model could include imitation learning or supervised learning to better reflect how dancers often learn from teachers. Another extension would be to add a generative component that allows the agent to create new choreography rather than reproduce a single fixed sequence. In that case, the reward function could evaluate broader choreographic principles such as movement continuity, rhythmic variation, motif repetition, symmetry, surprise, and diversity of movement. This would allow the model to move beyond memorization and toward a more realistic representation of dance as both a learned motor sequence and a creative art form.

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